Michael Xie (he/him) Electrical Engineering

Vancouver, BC | michaelxie04@gmail.com | +1 778-798-5779 | LinkedIn | Github | Available 4, 8, 12 months

TECHNICAL SKILLS

Electrical
- SMT and THT Soldering
- Multimeters, Oscilloscope, Power supply, waveform generator, breadboards
- Circuit Design/Components

Computer
- C++, MATLAB, Python, R
- SystemVerilog/Verilog, Quartus, ARM
- Arduino
- Visual Studio

Software
- Office Suite, G Suite
- GitHub
- SolidWorks, AutoCAD (inprogress)
- LaTeX
- Claude Code

EDUCATION

University of British Columbia - Okanagan **Expected Graduation: May 2027**
Bachelor of Applied Science - Electrical Engineering

Georgetown Business Academy, Washington **July 2020 – August 2021**
Summer Program

St. John's School, Vancouver **September 2016 – June 2022**
High School - Student Council Prefect

TECHNICAL WORK EXPERIENCE

University of British Columbia MEMS Lab, Vancouver, BC, Canada **April 2025 – June 2025**
Internship/Volunteer
- Assisted PHD student in cardiac tissue modelling using a MEMS-based cell stretcher and calibrated load cells for force measurement.
- Operated microscopes and collected substrate stress and strain data under various motor conditions
- Improved Python and C++ control code for ease of use and proposed hardware upgrades (e.g., closed-loop drivers, compact design, and magnetic placement).
- Enhanced clamping mechanism and enabled membrane underside observation for improved imaging.

HYC Technology, Suzhou, Jiangsu, China **July 2024 – August 2024**
Engineering & Operations Internship
- Completed rotational training across 7 departments including HR, supply chain, quality control, legal, semiconductor R&D, EV R&D, and smart device development.
- Assembled 100 smartwatch panel carriers for automated testing machines on the production line, gaining hands-on experience in manufacturing processes.
- Gained exposure to industry-standard tools such as Git, SolidWorks, Quartus Prime, Cadence Allegro, and Siemens/Mitsubishi PLC software.
- Explored internal systems including HYC's proprietary cloud network and remote desktop solutions for streamlined operations.

Suzhou Yi Li Energy, Suzhou, Jiangsu, China **July 2023 – August 2023**
Engineering Internship
- Conducted diagnostics and repairs on defective motors from the production line
- Utilized multimeters, oscilloscopes, and DC/AC generators to diagnose faulty PCBs and performed hands-on repairs by soldering and assembling components such as resistors, capacitors, and other critical components to restore PCB functionality.
- Gained insight into electric motor industry trends from a business and manufacturing perspective

LEADERSHIP

School Business Club, St. John's School **September 2020 – June 2022**
Team Leader
- Led student teams in business competitions, including the Wharton Global High School Investment Competition, achieving top placements.
- Conducted market research and analysis using tools such as the PEST (Political, Economic, Social, Technological) framework.
- Ranked top 1 in British Columbia and top 52 in Canada in the Junior Achievement Stock Market Competition; presented investment strategies to competition organizers and stakeholders.
- Developed leadership, strategic planning, and investment analysis skills through collaborative competition-based projects.

Project Plum Blossom, St. John's School **September 2020 – June 2021**
Co-Founder
- Led a team of three to launch a Stop Anti-Asian Hate initiative promoting cultural awareness and community support.
- Designed enamel pins and poster advertisements; coordinated with international manufacturers for production and logistics.
- Created and managed an Instagram campaign for national outreach, combining social media advertising with in-person sales at school events.
- Raised $1,123 for the Vancouver Chinatown Foundation to support elderly residents through both digital and in-person fundraising efforts.

Scouts Canada, Vancouver **September 2011 – June 2022**
Secretary, Venturer
- Planned events, led youth activities, maintained records, and promoted teamwork and safety during camps and community service projects.

TECHNICAL PROJECTS

Codling Moth Exuvia Counting System, University of British Columbia **September 2025- June 2026**
Client: Okanagan Kootenay Sterile Insect Release Program
- Delivered an automated exuvia counting system to an industry client, the Okanagan Kootenay Sterile Insect Release Program (OKSIR), as part of a 5-person team
- Built an enclosed imaging box with controlled white/UV LED lighting, a Raspberry Pi with a 12MP HQ camera, and a Streamlit web GUI requiring 3 clicks and ~2 minutes of processing per tray
- Trained a YOLOv8 detection model on 15,000+ labeled images using supervised and semi-supervised learning, reaching ~92% mAP@50
- Met all client targets (9% detection error, 3% repeatability variance, 95%+ uptime) and presented the final results at the UBC Okanagan capstone presentation and an on-site demo at the Osoyoos facility

32-Bit Modular Processor Design, University of British Columbia **May 2026 - June 2026**
- Designed a 32-bit modular ALU supporting a 15-instruction RISC architecture, engineered for standalone testability.
- Developed arithmetic modules, including a 32-bit adder, subtractor, and multiplier with custom carry-out and overflow flag logic (NZCV).
- Implemented parameterized shift and rotate registers (LSL, LSR, ROR) to execute dynamic data manipulation on source operands.
- Verified hardware functionality using ModelSim behavioural testbenches, ensuring timing accuracy and comprehensive edge-case coverage.

Brushless DC Motor, University of British Columbia **January 2025 – April 2025**
Assistant Captain
- Created a brushless DC motor with directional control and direction indicator from scratch
- Used SolidWorks to 3D print the chassis, then attached neodymium magnets, bearings, and copper wire
- Used Arduino to code directional, speed, and indicator controls
- Assistant group captain of the team of 6 to attain a grade 14/15

Self - Driving Crack Avoidance Pi-Car, University of British Columbia **January 2024 – April 2024**
Captain

- Designed and assembled an autonomous vehicle using motors, a camera, and a Raspberry Pi for real-time crack detection
- Programmed a convolutional neural network (CNN) in Python to process camera input and halt movement upon detecting surface cracks
- Trained the model using a dataset of 1,000 labelled images; fine-tuned architecture, activation functions, and training epochs to minimize prediction error
- Achieved a 98% success rate in accurately identifying and responding to cracks during testing

F-117A Nighthawk - SolidWorks Project, University of British Columbia **January 2023 – April 2023**
Captain

- Sketched and modelled the F-117A Nighthawk jet body in SolidWorks over 30 hours of design time.
- Assembled a full 3D jet model composed of ~5,000 parts, 200 subassemblies, and 1,500 mates, including wings, engine, fuselage, and landing gear.
- Created an animated simulation demonstrating mechanical movement of key jet components.
- Led a team of five as project captain, managing workflow, design integration, and final deliverables; earned an A grade on the project.

PROFESSIONAL AFFILIATIONS

Engineers and Geoscientists British Columbia (EGBC) **2024 – Present**

INTERESTS & ACTIVITIES

Sports

- Golfing and travelling
- High school volleyball, soccer, basketball, badminton
- Bronze Medallion swimming certification

Activities

- Piano, since age 6
- PC and VR gaming, Custom keyboard building
- Independent stock trading